



NFC (Near Field Communication) is a revolutionary wireless technology that enables seamless and secure contactless interactions between devices, revolutionizing industries like payments, access control, and data sharing.

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How NFC Works

1

Proximity

NFC relies on short-range (up to 4 inches) wireless communication between compatible devices.

2

Data Exchange

When devices are tapped or brought close together, they can securely exchange information such as payment details, login credentials, or content.

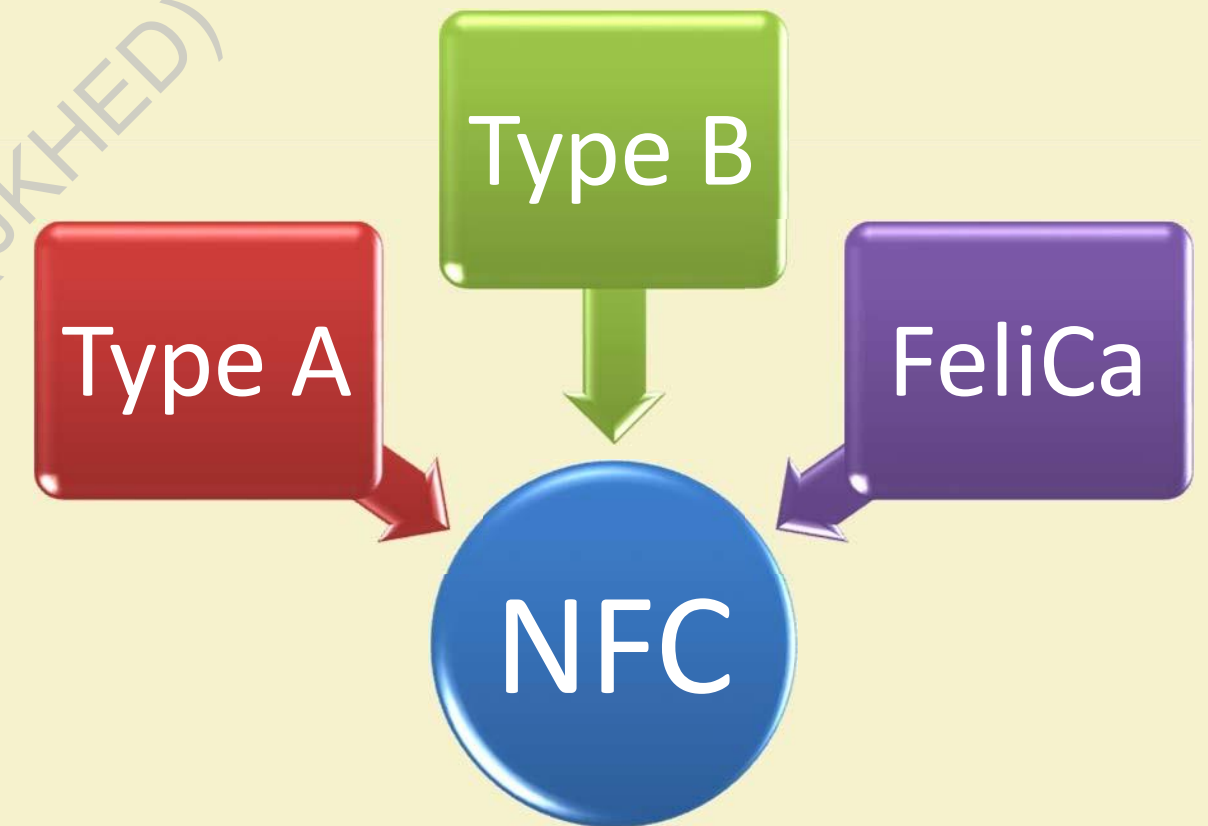
3

Simplicity

NFC is designed to be intuitive and user-friendly, requiring minimal setup or configuration for a seamless experience.

Introduction

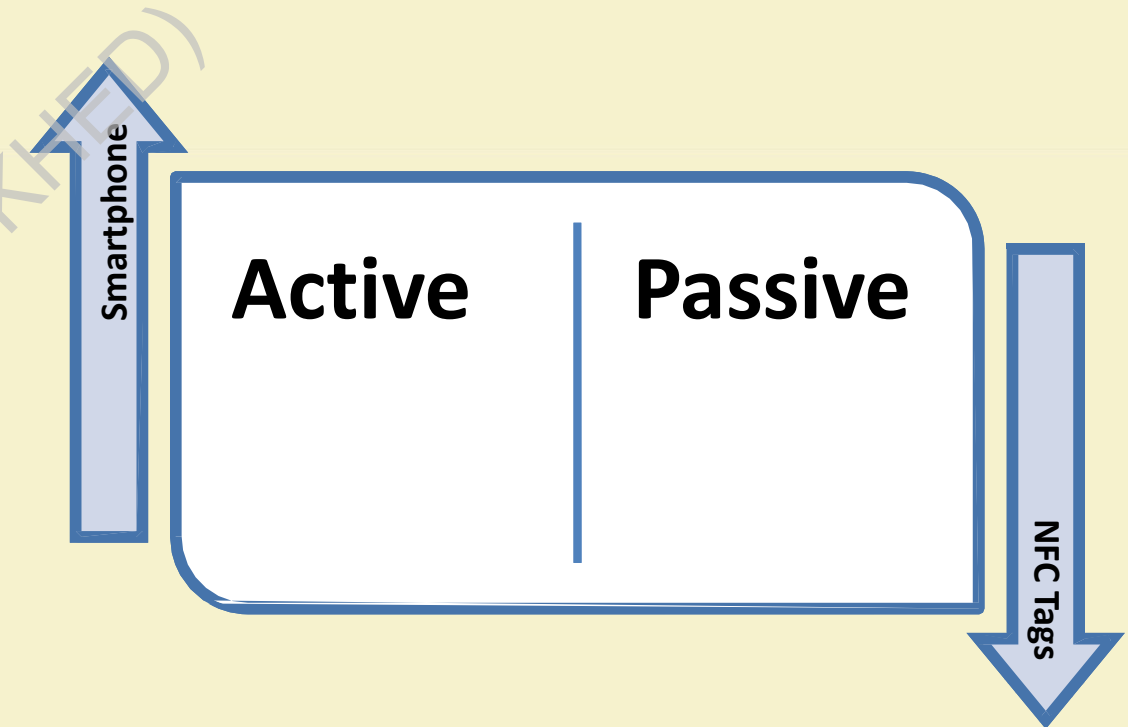
- ✓ **Near field communication**, or NFC for short, is an offshoot of radio-frequency identification (RFID).
- ✓ NFC is designed for use by devices within **close proximity** to each other.
- ✓ All NFC types are similar but communicate in slightly different ways.
- ✓ FeliCa is commonly found in Japan.



Source: "[How NFC Works](#)", NFC (Online)

NFC Types

- ✓ **Passive devices** contain information which is readable by other devices, however it cannot read information itself.
- ✓ NFC tags found in supermarket products are examples of passive NFC.
- ✓ **Active devices** are able to collect as well as transmit information.
- ✓ Smartphones are a good example of active devices.

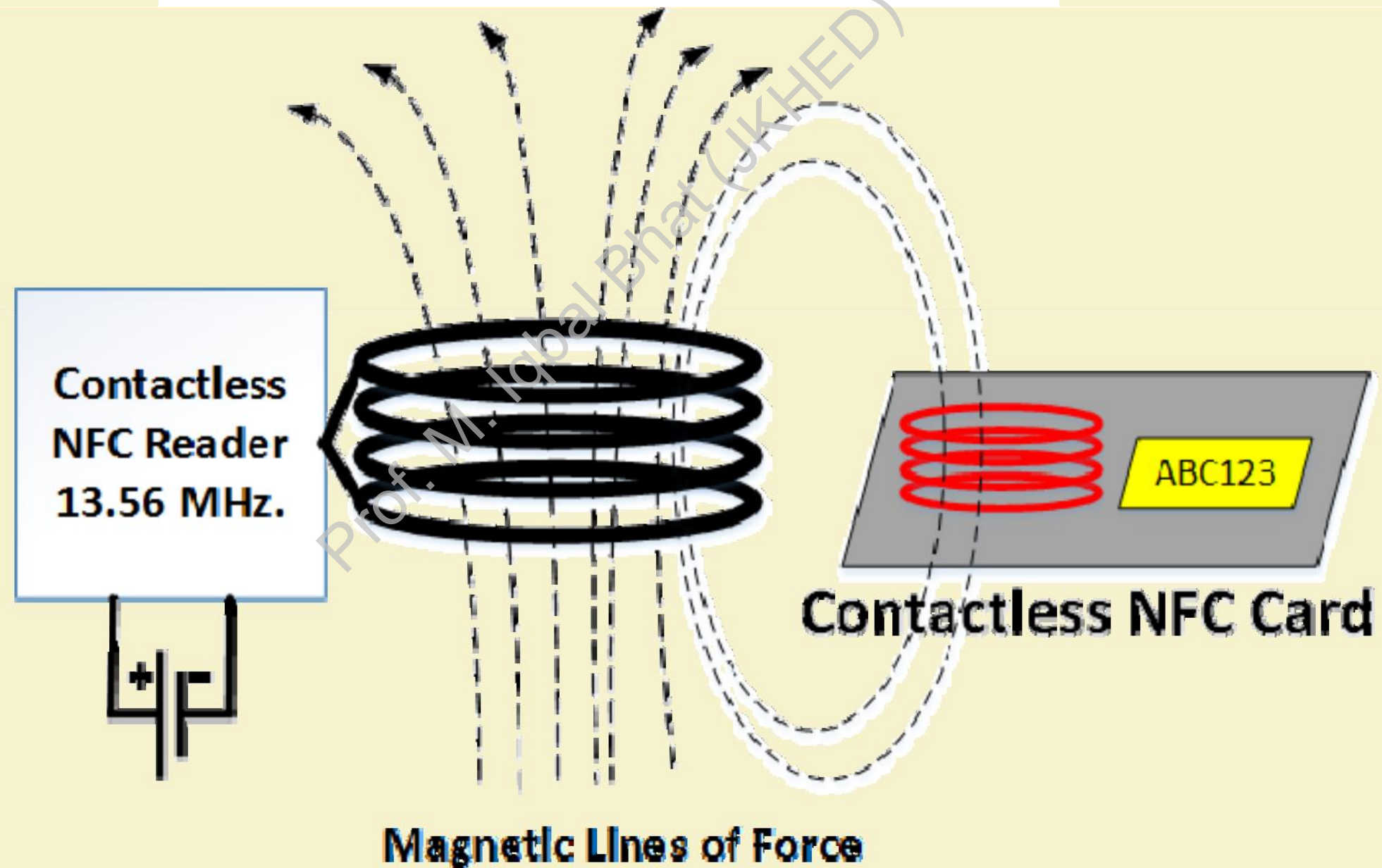


Source: "[How NFC Works](#)", NFC (Online)

Working Principle

- ✓ Works on the principle of **magnetic induction**.
- ✓ A reader emits a small electric current which creates a magnetic field that in turn bridges the physical space between the devices.
- ✓ The generated field is received by a similar coil in the client device where it is turned back into electrical impulses to communicate data such as identification number status information or any other information.
- ✓ 'Passive' NFC tags use the energy from the reader to encode their response while 'active' or 'peer-to-peer' tags have their own power source.

Source: "[Inside NFC: how near field communication works](#)", APC (Online), Aug. 2011



NFC Specifications

- ✓ NFC's data-transmission frequency is 13.56MHz.
- ✓ NFC can transmit data at a rate of either 106, 212 or 424 Kbps (kilobits per second).
- ✓ Tags typically store between 96 and 512 bytes of data.
- ✓ Communication range is less than 20cms.

Source: "[Inside NFC: how near field communication works](#)", APC (Online), Aug. 2011

Modes of Operation

Peer-to-peer

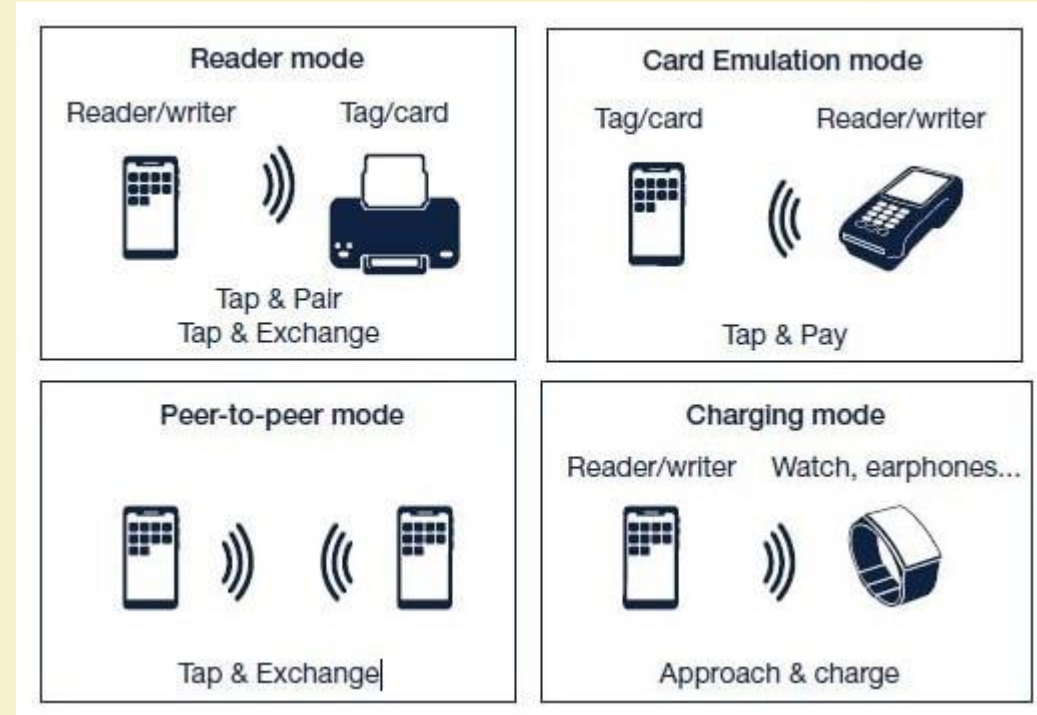
Lets two smartphones swap data

Read/Write

One active device picks up info from a passive one

Card emulation

NFC device can be used like a contactless credit card



Advantages of NFC Technology



Convenience

NFC provides a seamless and intuitive user experience, requiring minimal setup or configuration.



Security

NFC communication is encrypted and tokenized, ensuring a high level of data protection.



Speed

NFC-based transactions are fast and efficient, enabling quick and easy interactions.



Energy-Efficient

NFC requires minimal power consumption, making it a suitable choice for battery-powered devices.

NFC Security and Privacy

1 Encryption

NFC communication is encrypted, providing a high level of security to protect sensitive data.

3 User Control

Users have control over NFC transactions, requiring manual interaction to initiate data transfer or payments.

2 Tokenization

NFC payments use tokenization to replace sensitive card details with a unique identifier, further enhancing security.

4 Privacy

NFC's short-range communication limits the risk of unauthorized access or eavesdropping, preserving user privacy.

NFC Adoption and Trends

Global Growth

NFC technology is experiencing rapid global adoption, with an increasing number of devices and services supporting it.

Mobile Integration

Smartphones have become the primary NFC-enabled devices, driving the technology's widespread integration into everyday life.

Emerging Applications

NFC is finding new applications in areas like public transportation, access control, and data sharing, expanding its reach.

Ecosystem Expansion

A growing ecosystem of NFC-compatible devices, services, and platforms is further driving the technology's adoption and innovation.

NFC Applications

- ✓ Smartphone based payments.
- ✓ Parcel tracking.
- ✓ Information tags in posters and advertisements.
- ✓ Computer game synchronized toys.
- ✓ Low-power home automation systems.



